

Assignment: Numerical Integration

Name/ID: 유채은 / 22200476

Date: 2025 / 10 / 12

Academic Integrity Declaration

Check the statement that applies, and sign your name.

Check	Statement
☒	I have NOT used LLM (ChatGPT, etc), any online materials, or other students' reports/codes to complete this assignment.
☐	I have NOT completed this assignment entirely on my own.

If you received any assistance **that violates the class policy**, you must explain it in detail. A penalty will be applied depending on the **extent and nature of the assistance**.
Failure to disclose the assistance, you will get Grade F.

Examples:

- *Screenshots of LLM prompt,*
- *Explanation of assistance : who/how/what extent etc*

By signing below, I confirm that I have honestly and truthfully answered the academic integrity declaration.

Date: 2025.10.12

Sign: 유채은

Instruction

What you need to submit

Submit as a zip file: [Assignment_Integration_YuChaeun_22200476.zip](#) *

- **Report** (pdf)
- **Src Codes:** (1) [Assignment_Integration_YuChaeun_22200476.cpp](#)
(2) [myNP_22200476.h](#), [myNP_22200476.cpp](#)

* Change to your name and ID

Problem 1

Problem Statement

Use these discrete points for debugging.

```
x=[0 5 10 15 20 25 30 35 40 45 50 55 60];
y=[0 3 8 20 33 42 40 48 60 12 8 4 3];
I= 1397.500000 (Matlab output)
```

1. Create Simpson13() for discrete data inputs.

`double simpson13 (double x[ ], double y[ ], int m)`

- # subintervals= N, # intervals= N/2, # dataset=m=N+1
- N even numbers, same interval h
- This must be defined in **myNM.cpp** source file

Pseudocode

```
N = m - 1, h = x[1] - x[0]

// midpoints
    for k = 1 : (N / 2)
        I = I + h/3( 4y(2k - 1) + 2y(2k) )
    end
// first & last point + last 4f(x)
I = I + (h / 3) ( y(0) + 4 y(N - 1) + y(N) )

return I
```

Output window

```
// show your output
```

```
I_simpson13 = 1361.666667
```

2. Create Simpson38() for discrete data inputs.

`double simpson38 (double x[], double y[], int m)`

- # subintervals= N, # intervals= N/3, # dataset=m=N+1
- N multiple of 3, same interval h
- This must be defined in **myNM.cpp** source file

Pseudocode

```
// initialization
N = m - 1; // N=12
h = x[1] - x[0];

// midpoints
for k = 1 : (N / 3)
    I = I + (3h / 8) * ( 3y(3k-2) + 3y(3(k-1) + 2y(3k) );
end
// first & last point + last 4f(x)
I = I + (3h / 8) * ( y(0) + 3y(N - 2) + 3y(N - 1) + y(N) );

return I;
```

Output window

```
// show your output
```

```
I_simpson38 = 1434.375000
```

3. Create a numerical integration function when a mathematical function is given as the input.

- Create a function "myFunc()" to define the following integration. It should be defined in the **main** source file.

```
double myFunc (const double x) // in main.cpp
```

$$I(f) = \int_{-1}^1 \sqrt{1-x^2} dx \quad // \quad \text{It should be } \frac{\pi}{2}$$

- Use both Simpson13() and Simpson38() method depending on the value of N
- Choose different N: N=12, N=13

```
double integral(double func(const double x), double a, double b, int n)
```

Pseudocode

```
interval N
h = (b - a) / n;

midN = n - 3; // =====> must be even #

if (N == 2의 배수) { /* ----- simpson13 ----- */
    // midpoints
    for k = 1 : (N / 2)
        I = I + (h / 3) * (4func(a + (2k - 1)h) + 2func(a + 2hk));
    end

    // first & last point + last 4f(x)
    I = I + (h / 3) * (func(a) + 4func(b - h) + func(b));
}
}
```

```

else if (N == 3의 배수) { /* ----- simpson38 ----- */
    // midpoints
    for k = 1 : (N / 3)
        I = I + (3h / 8) * ( 3func(a + (3k - 2)h) + 3func(a + (3k - 1)h) +
2func(a + (3k)h));
    end
    // first & last point + last 4f(x)
    I = I + (3h / 8) * ( func(a) + 3 * func(b - 2h) + 3func(b - h) + func(b) );
else {
    /* ----- simpson13 for most points -----*/
    // midpoints
    for k = 1 : (midN / 2)
        I = I + (h / 3) * ( 4func(a + (2k - 1)h) + 2func(a + 2hk));
    end
    // first & last point + last 4f(x)
    I = I + (h / 3) * ( func(a) + 4func((a + midN*h) - h) + func(a + midN*h));

    /* ----- simpson38 for last three points ----- */
    I = I + (3h / 8) * ( func(b - 3h) + 3func(b - 2h) + 3 * func(b - h) + func(b));

return I;

```

Output window

```
// show your output
```

```

N = 12
I_function = 1.555063
Press any key to continue . . .

```

```

N = 13
I_function = 1.555333
Press any key to continue . . . |

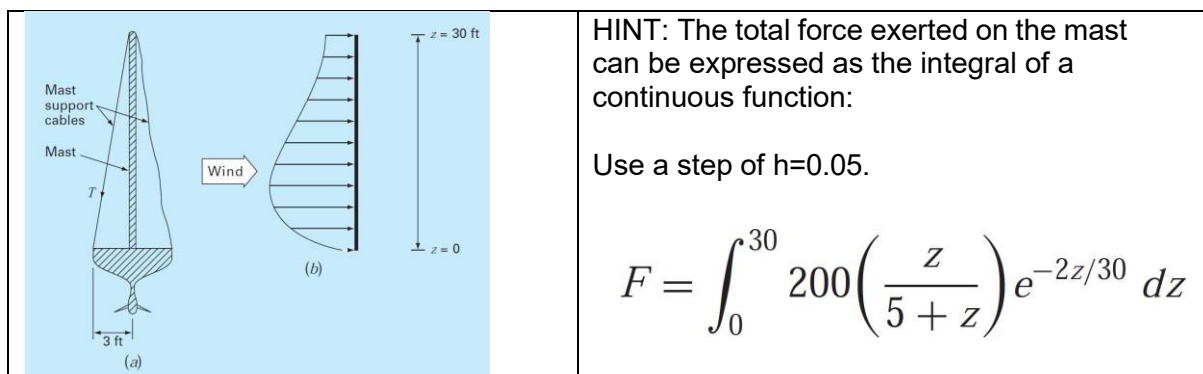
```

Problem 2:

Solve the following engineering problem with your library.

Across section of a racing sailboat is shown in below. Wind forces (f) exerted per foot of mast from the sails vary as a function of distance above the deck of the boat (z).

Calculate the tensile force T in the left mast support cable, assuming that the right support cable is completely slack and the mast joins the deck in a manner that transmits horizontal or vertical forces but no moments. Assume that the mast remains vertical.



ANS: 6.47kN